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IntelliBike: An Intelligent Bicycle with Automatic Shifting Based on Sensing and Riding Analysis

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***Abstract*— This paper presents an IntelliBike with automatic shifting based on the riding condition. It is able to collect and transmit multiple sensor data to the cloud database, and the real-time sensor information is used to analyze for an appropriate riding state through random forest. Our technique performs automatic gear shifting using inexpensive sensor components and embedded system development board, which is not available in current expensive electronic shifting systems. The function and the installation of the external components can be carried out on the off-the-shelf bicycle with a shifting line, without the need to dismantle or modify the basic equipment. The system also includes an IntelliBike-specific online management interface that allows all cloud data to be displayed and let users access and understand their riding conditions.**

I. INTRODUCTION

With the advances of modern technologies, in addition to bringing more convenience to our lives, it has also led to many environmental problems. Nowadays, most vehicles use gasoline or diesel as the main fuel, and the exhaust gas generated by combustion causes serious air pollution. People's awareness of environmental protection is getting stronger, and the idea of green transportation is now flooding the world. Riding a bicycle cannot only save energy and protect the environment, but also solve the traffic congestion problem in the metropolitan areas. To make bicycles easier to ride under different road and physical conditions, they are usually equipped with a gear shifting system. However, riding with incorrect gear might result in longer travel time, unnecessary physical strength waste, and even some physical injury to the body. Thus, the idea of automatic and intelligent shifting appears to be a feasible solution.

For the automatic gear change implementation, the timing of shifting is determined by a variety of sensors and microcontrollers. It allows the rider to have a better riding experience and concentrate on maintaining the balance of the bicycle. This can significantly reduce the accidents caused by the rider's distraction in adjusting the gear position. At present, the price of electronic transmission systems on the market is very expensive. It still needs manual control to change the gear, instead of automatic shifting according to the rider's state or the speed of the bicycle.

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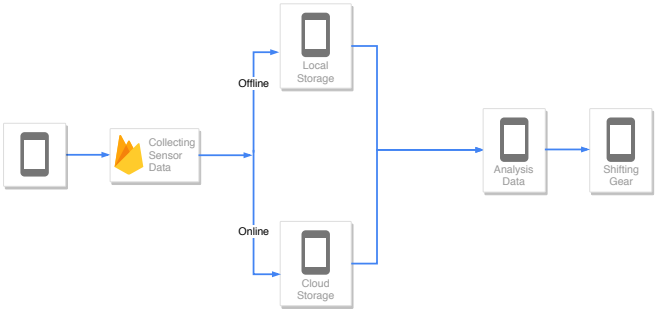

Fig. 1. A schematic diagram of the proposed IntelliBike system with automatic and intelligent gear shifting based on sensor data acquisition and analysis.

In this paper, we develop an IntelliBike with automatic shifting system. A schematic diagram is illustrated in Fig. 1. It takes the riding information such as moving speed, pedaling pressure, rider's heartbeat and slope of terrain, and uploads to the cloud database. After the data acquired by the sensors are collected, an analysis is carried out to calculate the shift timing for the user at each shifting stage. An automatic shifting operation is then performed accordingly. In our proposed IntelliBike, it is not necessary to think about the gear shifting behavior for every riding. The on-board microcontroller is used to analyze the most frequently changed gear of the rider using the data uploaded to the cloud, and take care of the motor powered gear shifting control. The experiments on the outdoor bicycle riding have demonstrated the feasibility of the proposed intelligent gear shifting technique.

II. RELATED WORK

In the previous work of automatic bicycle shifting systems, the relationship between the pedaling frequency and the wheel speed is considered as a key decision for gear shifting [21]. Alternatively, the pedal pressure and the wheel speed can be used to estimate the slope of the road, and the bicycle weight and load are taken into account to control the power of the electric motor [10]. One common point in the literature is using the relationship between the pedal and wheel speed to change gear or increase the motor power. However, the lacking of the physiological information of the human body and the actual slope of the road does not provide sufficient in-situ information for intelligent riding.

In [2], [6], the data from GPS and atmospheric pressure sensor are used to estimate the slope of the road. The slope is defined as the ratio of altitude change and distance traveled

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in a given time interval. Since the sampling frequency of GPS is low and the signals are easily blocked by obstacles such as tall buildings or forest, the slope estimation of the road can possess large errors due to the GPS signal delay. Furthermore, the atmospheric pressure sensor is affected by multiple factors such as temperature and wind speed, weather, etc., These factors will lead to an even larger computation error of the road slope. In this work, we use IMU combined with Kalman filter for sensor fusion and noise reduction to estimate the bicycle dynamic state [11].

In the past few years, different sensor fusion studies through Kalman filter have shown significant progress and prominent effects. Ferdinando *et al.* used Kalman filter for the fusion of gyroscope and accelerometer data to create a two-wheel balance robot based on ATMega8535 [5]. Gui *et al.* adopted a complementary filter and a Kalman filter to fuse the accelerometer and gyroscope data for tilt angle measurement on a MEMS based IMU [7]. Koksal *et al.* evaluated the sensor fusion techniques using Kalman filtering and complementary filtering for optimal attitude tracking of a quadcopter [13]. The results showed that the sensor fusion approach based on Kalman filter can provide minimum mean-squared-error estimates, as well as better attitude estimation and attitude control performance. Some recent works on sensor fusion through Kalman filter and IMU can be found in [4], [8], [12], [14], [16], [19], [20], [22]. These studies also include the extended Kalman filter (EKF), which illustrates the reliability of the IMU combined with Kalman filter. Zhang *et al.* proposed a positioning algorithm based on the GPS data and distributed Kalman filter fusion feedback. It combines the information from three GPS receivers to estimate the position, and solve the single-point GPS positioning error problem [23].

To collect the sensor data, Alzahranir introduced the use of Arduino or Raspberry Pi, combining a variety of different sensor systems to make IoT (Internet of Things) applications [1]. Both Arduino and Raspberry Pi are low-cost microcontrollers, and they are adopted by this work to make a bicycle IoT system. Arduino is used to collect the sensor data, which are then transmitted to the cloud database via Raspberry Pi.

For IoT related data analysis, Jing *et al.* used BPNN (Backpropagation Neural Network) and SVM (Support Vector Machine) to derive the relationship between AOD (Aerosol Optical Depth) and PM2.5/PM10 concentration [9]. It is then used to analyze and predict the ground air quality. The paper also reported that SVM had a better model prediction accuracy than BPNN. Patel *et al.* compared four different models, and analyzed and predicted data from the same stock market [18]. They used SVM, ANN (Artificial Neural Network), RF (Random Forest) and NB (Naïve Bayes) as training and testing models separately. The comparison results show that the best strategy for discrete stock market data forecasting is to use the RF decision model, and ANN stands for the second choice. For mobile applications, Cao *et al.* used RF to classify the driving behavior of motorists based on three-axis accelerometers, and compared them with other machine learning methods, including SVM, NB, K-means, DT (Decision-Tree) and EM clustering (Expectation-Maximization) [3]. RF has a maximum accuracy of 98.1%, and the overall accuracy of adding two more complex driving behaviors is 94.7%, which verifies the robustness of the RN method.

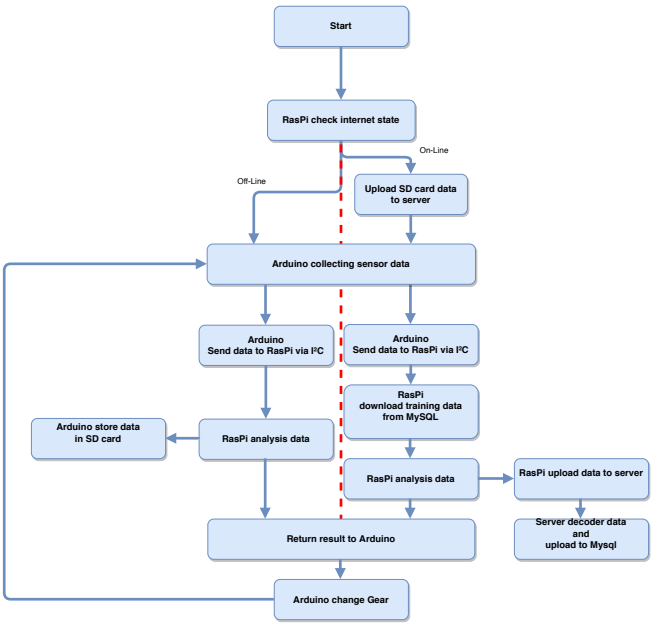

Fig. 2. The system flowchart of the proposed IntelliBike implementation. It first confirms if the network is connected via Raspberry Pi, and follows different processing paths.

The literature review shows that the application of RF has significant effects in various fields. Specifically, when new users access a system with a small amount of initial training materials, RF is suited for decision making. If the architecture using neural network requires a large amount of training data, RF is able to balance the training efficiency and accuracy. Thus, in this work, RF is adopted as the data analysis method and decision model, and used to determine the timing of automatic gear shifting.

III. PROPOSED APPROACH

The flowchart of the proposed system is illustrated in Fig. 2. At the beginning, the system first confirms if the network is connected via Raspberry Pi. If it is offline, Arduino stores the information on the SD card after collecting the sensor data. The data are also transmitted to Raspberry Pi through I^2C (Inter-Integrated Circuit). Raspberry Pi uses the historical riding information downloaded from the cloud database at the last time connected to the network as the training data for gear shifting classification by random forest. The result is then passed back to Arduino via I^2C for gear shifting through the servo motor control.

If the state is online, the sensor data are directly transmitted to Raspberry Pi via I^2C after collected by Arduino. After the data are received by Raspberry Pi, they are uploaded to the server for text decoding, and then saved to MySQL cloud database. On the other hand, the historical riding records

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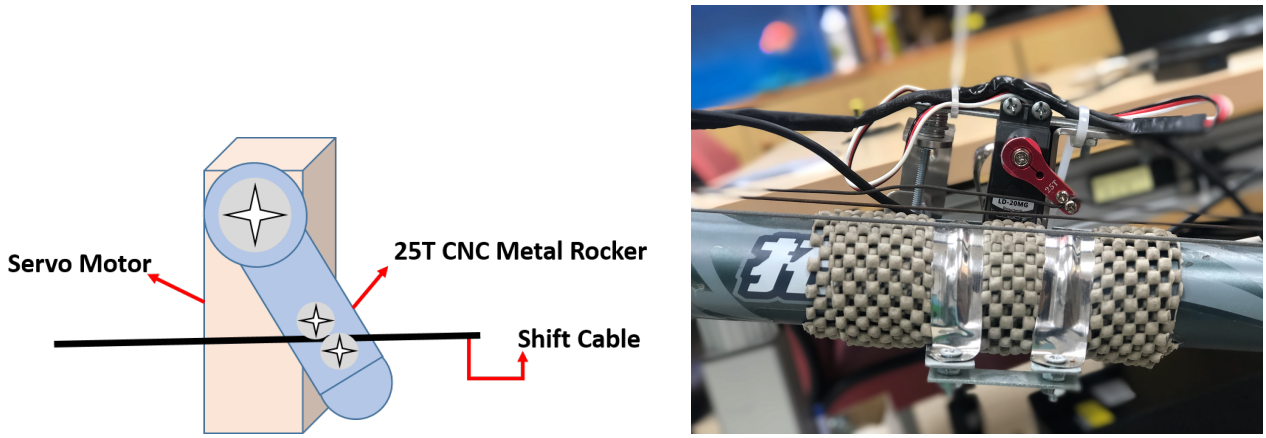

(a) The schematic diagram of our gear shifting module design with a servo motor, a 25T CNC rocker arm, and a shift cable.

(b) The actual implementation of the gear shifting module mounted on an off-the-shelf bicycle.

Fig. 3. The design and implementation of our gear shifting module.

will be downloaded from MySQL cloud database as the training data for RF. The on-site sensing information is then used to make the gear shifting decision by RF, followed by the operation executed by Arduino via I^2C for gear shifting through the motor control.

Our designed gear shifting module is shown in Fig. 3(a). It consists of a servo motor, a 25T CNC rocker arm, and a shift cable. The shift cable is placed in the center of the two screw holes above the rocker arm and tightened with the screws so it can be fixed above the rocker arm without slipping. Finally, the gear shifting module is mounted on the bicycle frame to convert the conventional manual shifting system into an electronic shifting system in the simplest and cheapest way. It is installed without replacing the entire shifting system with a complete automatic transmission product. The rotation of the servo motor drives the shift cable, which causes the bicycle's original shifting mechanism to adjust the chain for shifting. Fig. 3(b) shows the gear shifting module mounted on a bicycle.

Two microcontrollers, Arduino MEGA 2560 R3 and Raspberry Pi 3, and a number of sensors are used in the hardware development. The sensor modules adopted in this work consist of the followings: hall effect sensor, finger heartbeat sensor, pedal pressure sensor, six-axis IMU (MPU6050), real-time clock, GPS (Ublox NEO-6M V2), servo motor, I^2C multiplexer (TCA9548A), and SD card. We use Arduino as the information center and local storage base, and Raspberry Pi as the decision center and internet connecting point. These two microcontrollers communicate via I^2C.

For the software development, Arduino IDE is used for Arduino coding. Programs written on Arduino IDE are often called Sketches and saved as the .ino extension. Python is used for Raspberry Pi coding for the I^2C receiver and transmitter, and the internet client. The PC server uses C/C++ for socket coding to receive the client data and decoding for the receiver buffer. The server will upload the data to MySQL database after decoding the receiver buffer. We use PHP, HTML, JavaScript and CSS to create our IntelliBike website[1]. Fig. 4 illustrates the design of online management interface. It includes an OSM (Open Street Map) and a riding information chart. The riding information consists of heart rate, pedal pressure, slope of the road, riding speed, gear. The objective of the chart design is to help the user understand the current and historical riding information.

IV. EXPERIMENTS

In this work, we use IMU combined with Kalman filter for sensor fusion and reducing noise interference to estimate the bicycle dynamic state. Kalman filter generally needs a high sampling rate, but the adopted processing unit, Arduino Mega 2560 R3, only has single core and cannot use multi-thread to implement the high sampling rate sensor fusion operation. Thus, the multi-tasking, as the flowchart shown in Fig. 5, is adopted for this purpose. When the main program is running, an interrupt signal is sent out every 10 milliseconds. The program accesses the IMU for sensor data acquisition and performs the operations of Kalman filter per 10 milliseconds. After the interrupt program is executed, the main program of the original system continues. This method can be used to obtain the dense IMU information for Kalman filter.

The result comparison of the original IMU data and processed with Kalman filter is shown in Fig. 6. The test actions in the figure are immobility $\rightarrow$ tap the handle of bicycle $\rightarrow$ immobility. The blue signal indicates the information of the IMU's Y-axis accelerometer converted to the tilt angle. The red signal shows the data fusion result of the Y-axis accelerometer and the angular value of the Z direction gyroscope after passing through Kalman filter and converting to the tilt angle.

In this paper, we use random forest to predict the gear shifting position of bicycle riding. It takes the inclination of the bicycle, riding speed, the pressure of the pedaling, and the heart rate as the inputs for prediction. According to the

[1]IntelliBike Website: http://vision.ee.ccu.edu.tw/intellibike/

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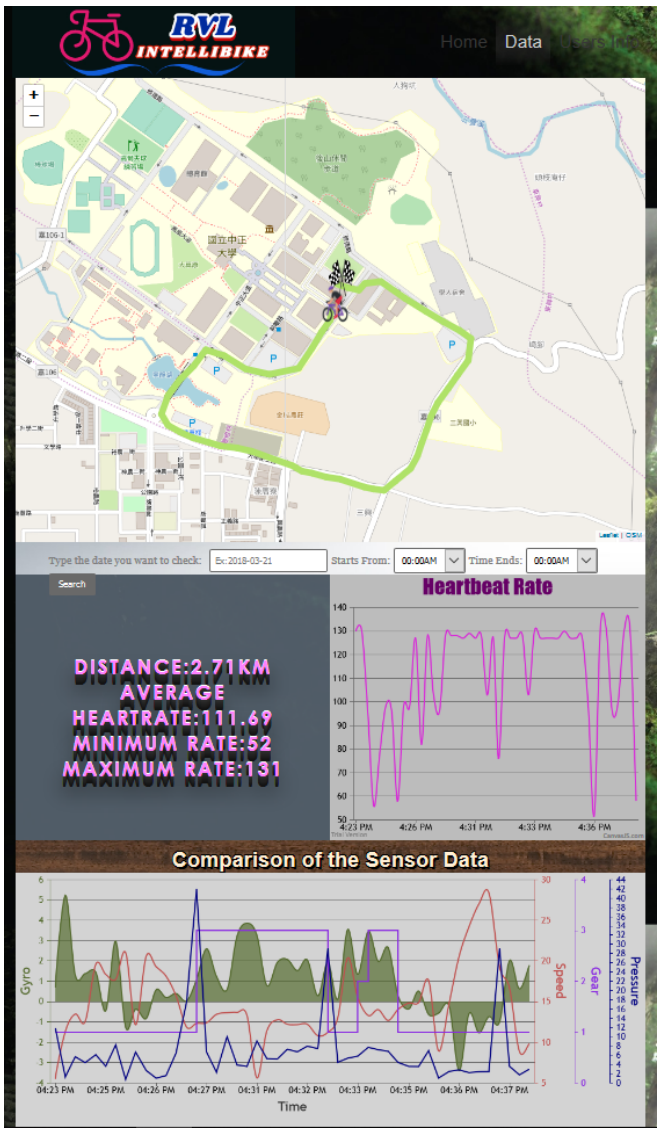

Fig. 4. The online management interface of our IntelliBike website. It can be accessed at http://vision.ee.ccu.edu.tw/intellibike/

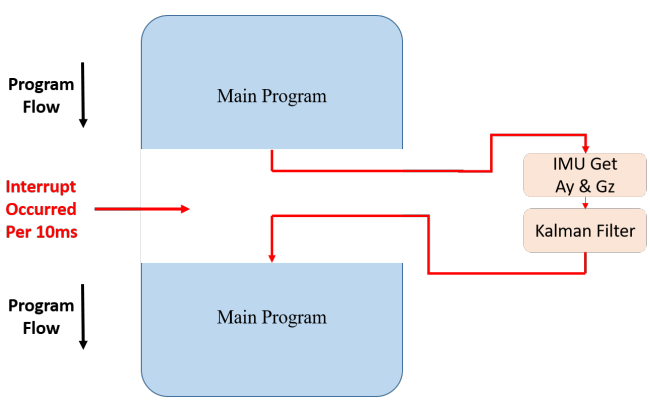

Fig. 5. The multi-tasking flowchart for dense IMU data access on Arduino.

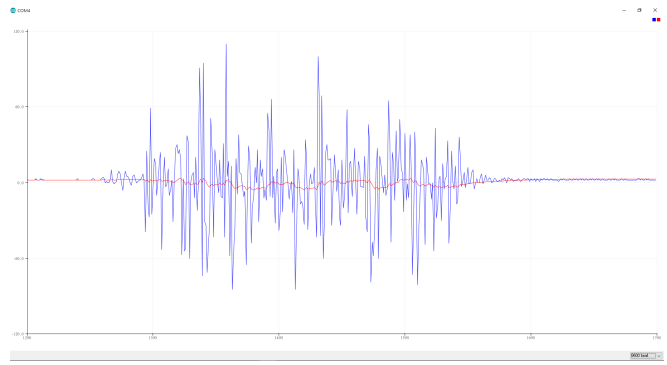

Fig. 6. The original IMU signal (in blue) and processed by Kalman filter (in red).

TABLE I

THE GROUND-TRUTHS OF THE TRAINING AND TESTING DATA.

Class of Gear	*Training Data*	*Testing Data*
Gear 1	81	25
Gear 2	16	0
Gear 3	15	8
Gear 4	19	1
Gear 5	3	2
Gear 6	28	10
Total	162	45

empirical analysis using random forest to forecast 29 different datasets in [17], the result shows that the accuracy of about 128 trees will be similar. Thus, the number of decision trees for gear shifting position prediction in this paper is set to 150. The total riding distance of the training data in Table I is approximately 8.34 km with 162 data inputs. The training data are generated by our own riding experience with manual gear shifting. The total riding distance of the testing data in Table I is approximately 2.12 km with 45 data inputs.

Fig. 7 shows the prediction result, with the ground-truth marked in green and the prediction marked in blue. Table II tabulates the results of predictions, with the accuracy of about 71.7%. The result also shows that only Gear 1 and Gear 6 have the high accuracy. This might be due to the use of subjective riding experience as the ground-truth, and the physical strength factors cause the prediction results not prominent for the middle range gears.

Since only the riding speed, pedaling pressure, heartbeat,

TABLE II

THE TESTING DATA GROUND-TRUTH AND THE TESTING RESULTS IN THE FIRST EXPERIMENT.

Class of Gear	*Testing Ground-Truth*	*Testing Result*
Gear 1	24	24
Gear 2	0	0
Gear 3	8	0
Gear 4	1	0
Gear 5	2	0
Gear 6	10	9
Total	45	33

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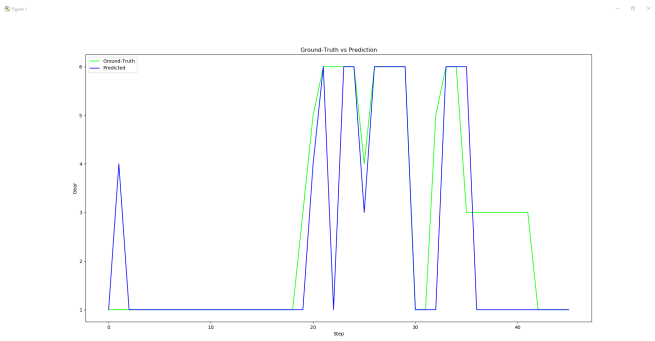

Fig. 7. The random forest prediction of gear shifting based on Gini impurity. The ground-truth curve is shown in green and the prediction curve is shown in blue.

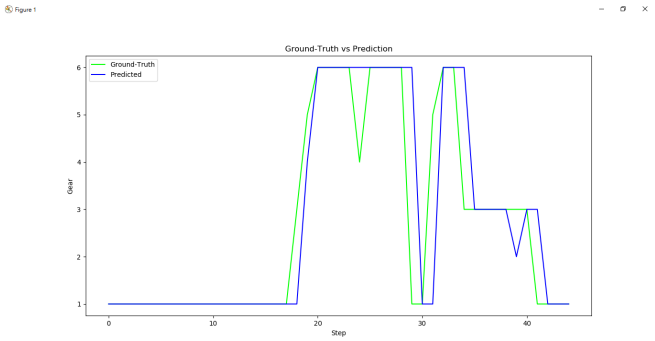

Fig. 8. The gear shifting prediction derived with the previous gear shifting information. The blue prediction curve shows an improvement compared to the result in Fig. 7.

and the inclination of the bicycle are used for prediction, it is very likely that when the physical strength of the cyclist is poor, the gear shifting information generated by random forest cannot meet the actual riding state. Thus, another experiment is carried out with previous gear shifting information, and the result is shown in Fig. 8. The prediction accuracy is greatly improved to an overall 82.2% as indicated in Table III.

Further examining the result curves in Fig. 8, it can be seen that the overall prediction seems delayed in time, although most of the classification have achieved a certain accuracy. This could be due to the ground-truth data collected by the cyclists and their expectation for early gear shifting when facing uphill or downhill. For instance, when the terrain in front is uphill, the action of an early shifting will be performed by the cyclist based on the expected psychological factor, even if the ground is still flat (and detected by the sensors). To eliminate this mismatch due to the inappropriate early shifting, the ground-truth data are moved forward by one displacement. Fig. 9 and Table IV show the results after the expected psychological factor is removed. The predict accuracy is then updated to 88.6%.

In the last experiment, we use a different route as the training and testing data to verify the proposed technique, and the training and testing ground-truths are shown in Table V. We perform three different methods using the testing data, and

TABLE III

THE TESTING DATA GROUND-TRUTH AND THE TESTING RESULTS WITH THE PREVIOUS GEAR SHIFTING INFORMATION PROVIDED.

Class of Gear	*Testing Ground-Truth*	*Testing Result*
Gear 1	24	22
Gear 2	0	0
Gear 3	8	5
Gear 4	1	0
Gear 5	2	0
Gear 6	10	10
Total	45	37

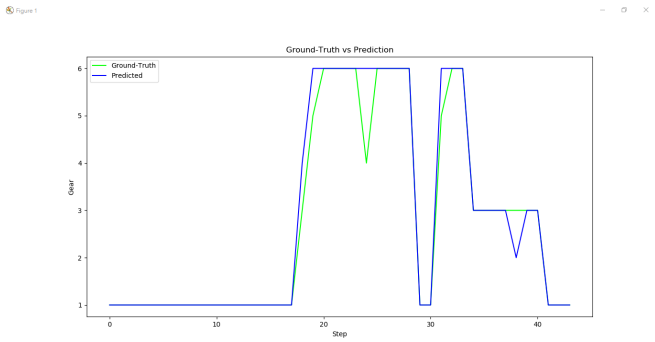

Fig. 9. The gear shifting prediction after the elimination of expected psychological factor. The blue prediction curve shows an improvement compared to the result in Fig. 8.

analyze the prediction accuracy. The first method uses the riding speed, pedaling pressure, heart rate and the inclination of the bicycle for prediction. The second method additionally add the previous gear shifting information for prediction. The third method further incorporates the elimination of the expected psychological factor to predict. The prediction accuracies of these three approaches are 68.7%, 89.3% and 91.3%, respectively. It is consistent with the previous experimental results that the high accuracy is obtained by considering the two additional factors.

Finally, we compare the computation load with the neuro-fuzzy technique [15] on a PC platform with Intel I7-7700 CPU. Table VI shows the comparison result of the execution time using the same training and testing data. 150 decision trees are used for random forest (both Gini impurity and entropy) and neuro-fuzzy is trained with 8 epochs. The result shows that the cost of neuro-fuzzy is too high, and it is not possible to adopt to our Raspberry Pi for online execution. In the empirical test, executing random forest on Raspberry Pi in our implementation takes about one second.

V. CONCLUSIONS

In this paper, we present the design and implementation of an intelligent bicycle with automatic gear shifting based on the riding condition. It consists of an automatic shifting mechanism which can be installed on the off-the-shelf bicycle easily. The automatic gear shifting is implemented using random forest based on the data collected from the environment sensing and the user's physiological information. In addition, an IntelliBike website is built to assist the

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TABLE IV

THE TESTING DATA GROUND-TRUTH AND THE TESTING RESULTS WITH THE ELIMINATION OF EXPECTED PSYCHOLOGICAL FACTOR.

Class of Gear	*Testing Ground-Truth*	*Testing Results*
Gear 1	23	23
Gear 2	0	0
Gear 3	8	6
Gear 4	1	0
Gear 5	2	0
Gear 6	10	10
Total	34	39

TABLE V

THE GROUND-TRUTHS OF THE TRAINING AND TESTING DATA.

Class of Gear	*Training Ground-Truth*	*Testing Ground-Truth*
Gear 1	59	31
Gear 2	8	1
Gear 3	21	16
Gear 4	5	0
Gear 5	5	0
Gear 6	10	0
Total	108	48

users to check their riding information and provide feedbacks for automatic gear shift learning. The proposed system can be integrated to a shared-bike network to acquire real-time data for online analysis. A better riding experience can be expected with more sensor data incorporated and less user decision on technological adaption.

ACKNOWLEDGMENT

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REFERENCES

[1] S. M. Alzahrani. Sensing for the internet of things and its applications. In *2017 5th International Conference on Future Internet of Things and Cloud Workshops (FiCloudW)*, pages 88–92, Aug 2017.

[2] H. S. Bae, J. Ryu, and J. C. Gerdes. Road grade and vehicle parameter estimation for longitudinal control using gps. In *Proceedings of the IEEE Conference on Intelligent Transportation Systems*, pages 25–29, 2001.

TABLE VI

THE COMPARISON OF EXECUTION TIME OF THREE DIFFERENT METHODS.

Method	*Computation Time (Second)*
Random Forest (Gini Impurity)	0.136
Random Forest (Entropy)	0.218
Neuro-Fuzzy	7725.601

[3] W. Cao, X. Lin, K. Zhang, Y. Dong, S. Huang, and L. Zhang. Poster abstract: Analysis and evaluation of driving behavior recognition based on a 3-axis accelerometer using a random forest approach. In *2017 16th ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*, pages 303–304, April 2017.

[4] S. Evren, F. Yavuz, and M. Unel. High precision stabilization of pan-tilt systems using reliable angular acceleration feedback from a master-slave kalman filter. *Journal of Intelligent & Robotic Systems*, 88(1):97–127, 2017.

[5] H. Ferdinando, H. Khoswanto, and D. Purwanto. Embedded kalman filter for inertial measurement unit (imu) on the atmega8535. In *2012 International Symposium on Innovations in Intelligent Systems and Applications*, pages 1–5. IEEE, 2012.

[6] R. Goussault, A. Chasse, and F. Lippens. Model based cyclist energy prediction. In *2017 IEEE 20th International Conference on Intelligent Transportation Systems (ITSC)*, pages 1–6, Oct 2017.

[7] P. Gui, L. Tang, and S. Mukhopadhyay. Mems based imu for tilting measurement: Comparison of complementary and kalman filter based data fusion. In *2015 IEEE 10th conference on Industrial Electronics and Applications (ICIEA)*, pages 2004–2009. IEEE, 2015.

[8] F. Hoffmann, N. Goddemeier, and T. Bertram. Attitude estimation and control of a quadrocopter. In *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 1072–1077. IEEE, 2010.

[9] L. Jing, X. Feng, L. Jintuo, M. Rui, L. Weiliang, and L. Yongjun. Contrastive research of svm and bp neural network in aod prediction. In *2017 36th Chinese Control Conference (CCC)*, pages 4072–4076, July 2017.

[10] Z. Kai and Y. Dejun. A control approach adaptive to load and road slope for electric power assisted bicycle. In *2017 36th Chinese Control Conference (CCC)*, pages 3414–3418. IEEE, 2017.

[11] R. E. Kalman. A new approach to linear filtering and prediction problems. *Journal of basic Engineering*, 82(1):35–45, 1960.

[12] A. Kim and M. Golnaraghi. A quaternion-based orientation estimation algorithm using an inertial measurement unit. In *PLANS 2004. Position Location and Navigation Symposium (IEEE Cat. No. 04CH37556)*, pages 268–272. IEEE, 2004.

[13] N. Koksal, M. Jalalmaab, and B. Fidan. Adaptive linear quadratic attitude tracking control of a quadrotor uav based on imu sensor data fusion. *Sensors*, 19(1):46, 2019.

[14] S. Lesecq, S. Gentil, and N. Daraoui. Quadrotor attitude estimation with data losses. In *2009 European Control Conference (ECC)*, pages 3851–3856. IEEE, 2009.

[15] C.-T. Lin, C. G. Lee, C.-T. Lin, and C. Lin. *Neural fuzzy systems: a neuro-fuzzy synergism to intelligent systems*, volume 205. Prentice hall PTR Upper Saddle River NJ, 1996.

[16] J. L. Marins, X. Yun, E. R. Bachmann, R. B. McGhee, and M. J. Zyda. An extended kalman filter for quaternion-based orientation estimation using marg sensors. In *Proceedings 2001 IEEE/RSJ International Conference on Intelligent Robots and Systems. Expanding the Societal Role of Robotics in the the Next Millennium (Cat. No. 01CH37180)*, volume 4, pages 2003–2011. IEEE, 2001.

[17] T. M. Oshiro, P. S. Perez, and J. A. Baranauskas. How many trees in a random forest? In *International workshop on machine learning and data mining in pattern recognition*, pages 154–168. Springer, 2012.

[18] J. Patel, S. Shah, P. Thakkar, and K. Kotecha. Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications*, 42(1):259–268, 2015.

[19] A. M. Sabatini. Quaternion-based extended kalman filter for determining orientation by inertial and magnetic sensing. *IEEE Transactions on Biomedical Engineering*, 53(7):1346–1356, 2006.

[20] A. M. Sabatini. Kalman-filter-based orientation determination using inertial/magnetic sensors: Observability analysis and performance evaluation. *Sensors*, 11(10):9182–9206, 2011.

[21] P. Tandon, A. Awasthi, B. K. Mishra, P. Rathore, and R. K. Shukla. Design and simulation of an intelligent bicycle transmission system. *IEEE/ASME Transactions on Mechatronics*, 16(3):509–517, 2011.

[22] S. Wang and Y. Yang. Quadrotor aircraft attitude estimation and control based on kalman filter. In *Proceedings of the 31st Chinese Control Conference*, pages 5634–5639. IEEE, 2012.

[23] M. Zhang, X. Jia, B. Yang, X. Chi, and D. Peng. A gps positioning algorithm based on distributed kalman filter data fusion with feedback. In *2018 37th Chinese Control Conference (CCC)*, pages 7359–7363. IEEE, 2018.

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